# BAB 1

## A. SISTEM KOORDINAT

Disadur dari Magdy Iskander, Electromagnetic fields and waves

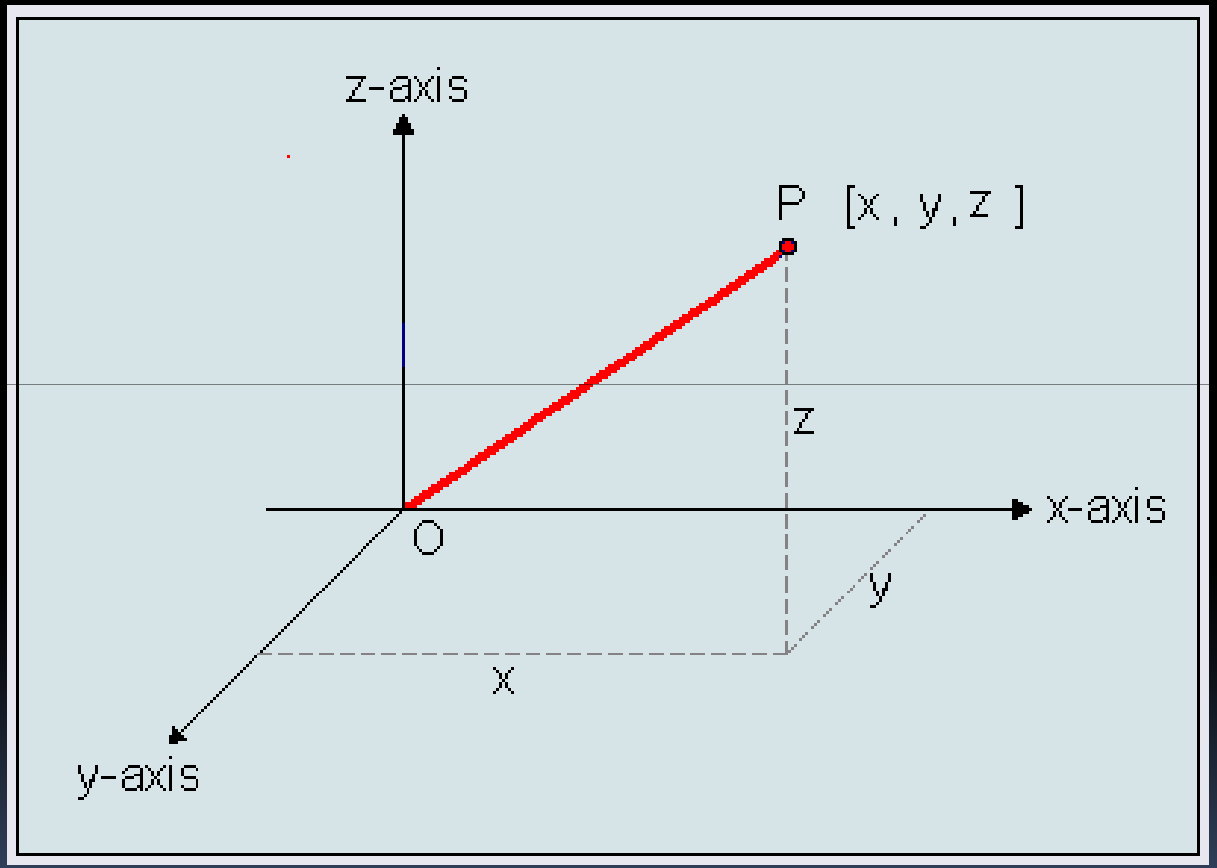
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# Sistem koordinat

- Sebuah titik ditentukan oleh perpotongan 3 buah bidang yang saling tegak lurus
- Suatu vektor pada sistemkoordinat dinyatakan dalam komponen komponen yang arahnya searah *vektor satuan*
- Vektor satuan adalah vektor dengan magnitude 1 dengan arah tegak lurus terhadap bidang koordinat konstant

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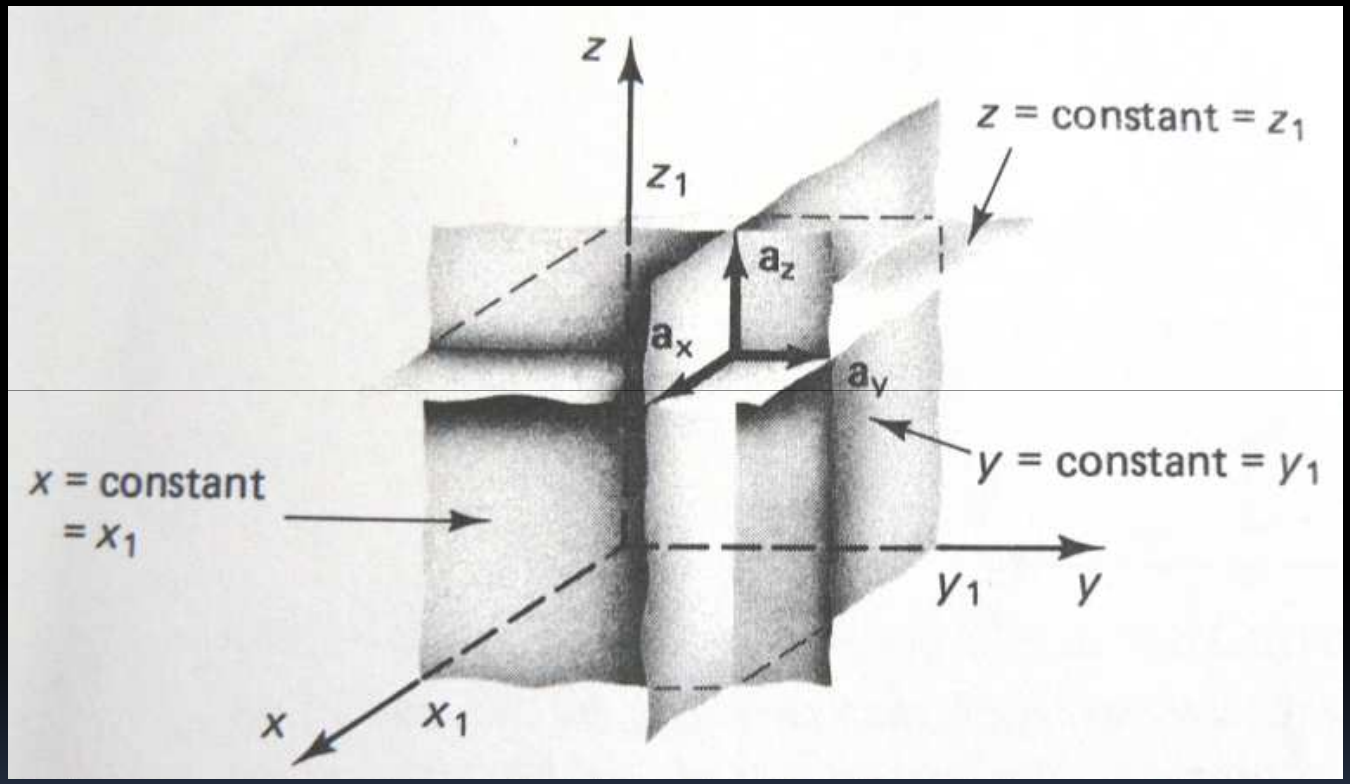
# Sistem koordinat Cartesian (1)




- Sebuah titik P dinyatakan oleh P(x, y, z)

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# Sistem koordinat Cartesian (2)

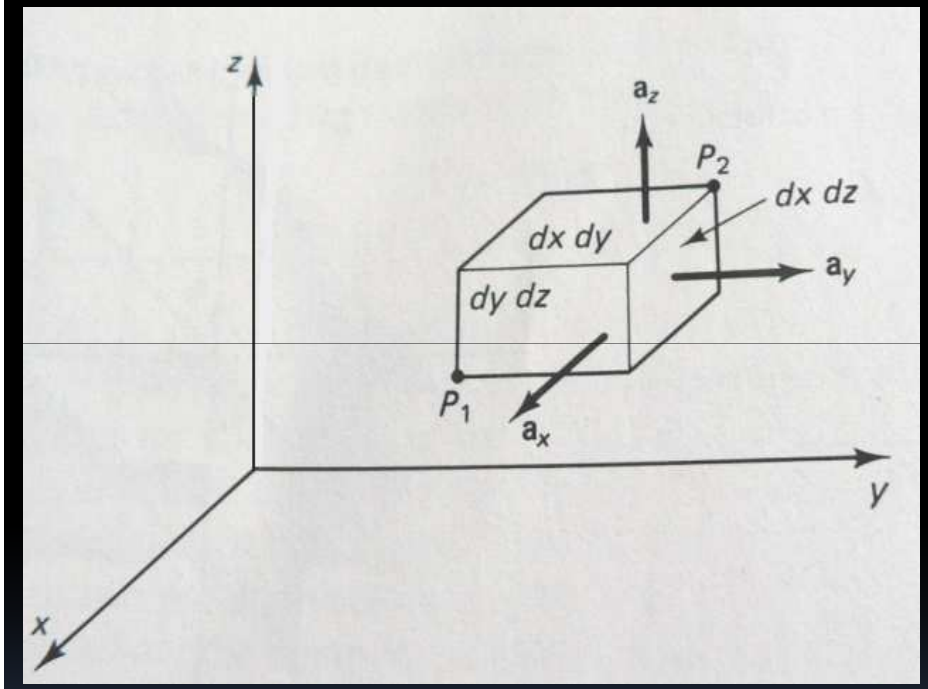



- Sebuah vektor **P** dinyatakan dengan :

$$\mathbf{P} = P_x \mathbf{a}_x + P_y \mathbf{a}_y + P_z \mathbf{a}_z$$

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# Sistem koordinat Cartesian (3)




- $P_1(x, y, z)$
- $P_2(x + \Delta x, y + \Delta y, z + \Delta z)$
- Panjang differensial d**l**

  $d\ell = dx\ \mathbf{a}_x + dy\ \mathbf{a}_y + dz\ \mathbf{a}_z$
- Luas differensial

  $d\mathbf{s}_x = dy\ dz\ \mathbf{a}_x$

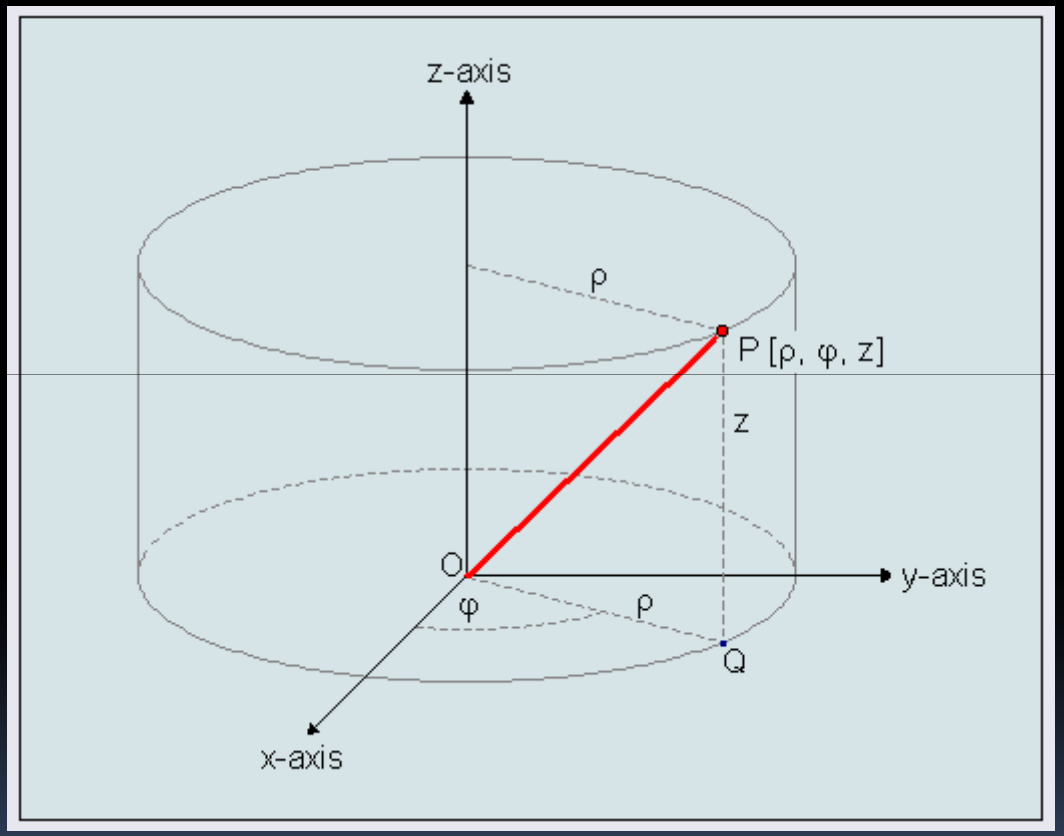
  $d\mathbf{s}_y = dx\ dz\ \mathbf{a}_y$

  $d\mathbf{s}_z = dx\ dy\ \mathbf{a}_z$
- Volume differensial

  $dv = dx\ dy\ dz$

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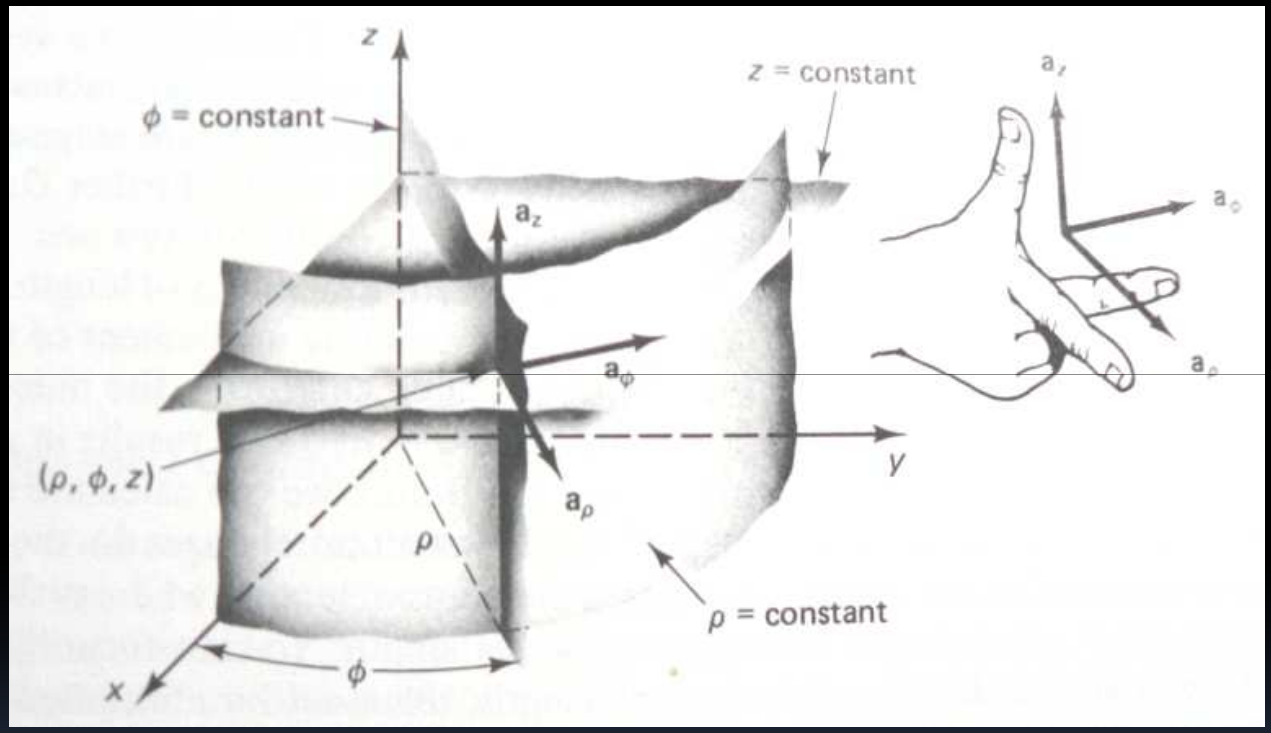
# Sistem koordinat silinder (1)




- Sebuah titik P dinyatakan oleh P(ρ, φ, z)

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# Sistem koordinat silinder (2)

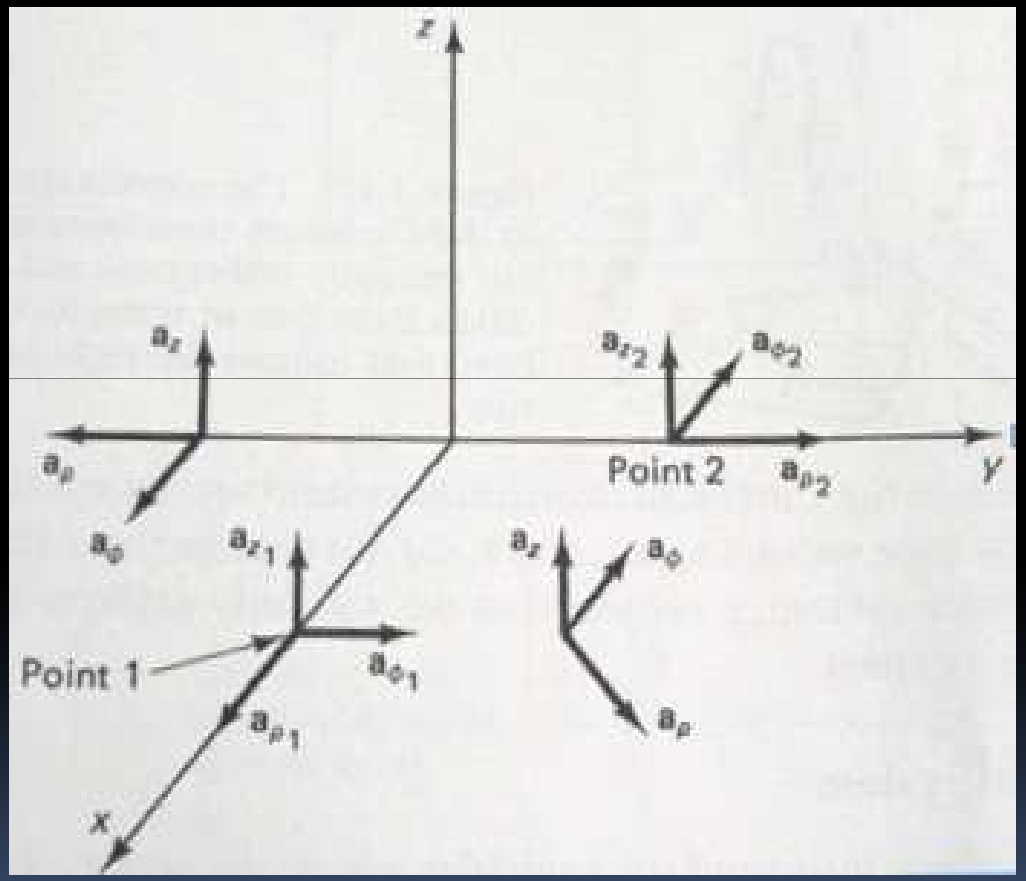



- Sebuah vektor **P** dinyatakan dengan :

$$\mathbf{P} = P_\rho \, \mathbf{a}_\rho + P_\phi \, \mathbf{a}_\phi + P_z \, \mathbf{a}_z$$

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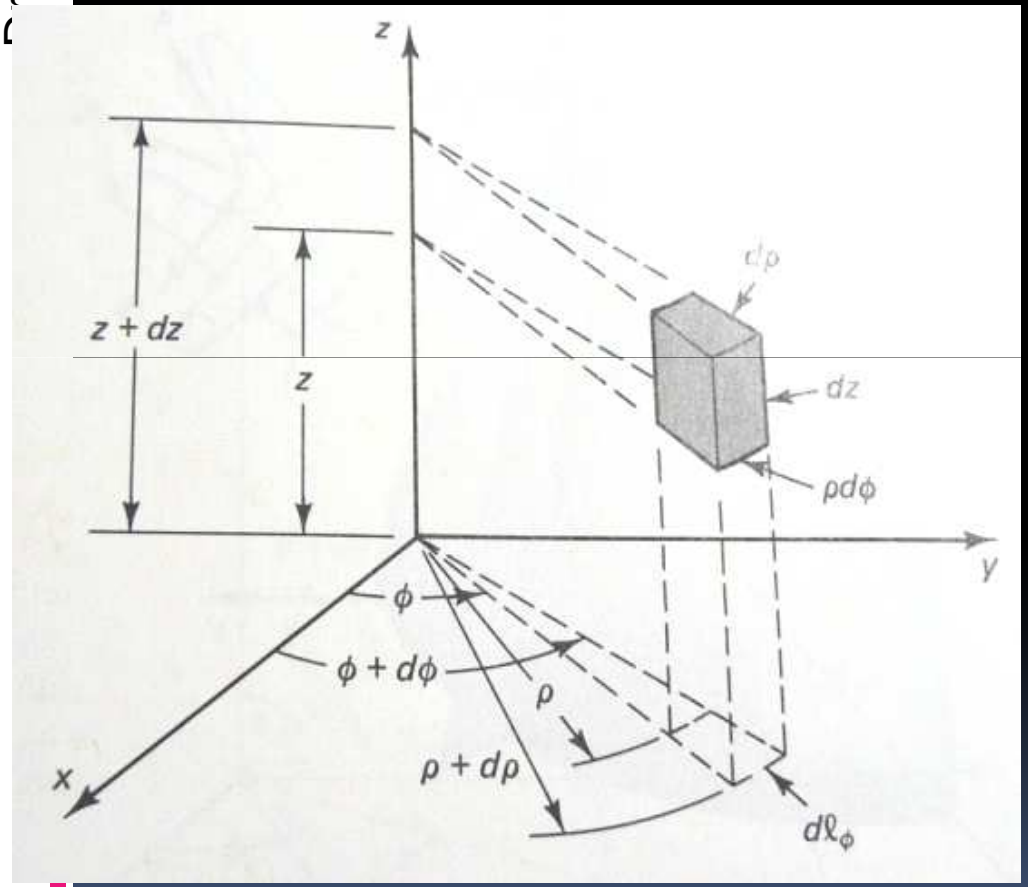
# Sistem koordinat silinder (3)




- Perhatikan arah vektor satuan yang berubah di tiap titik

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# Sistem koordinat silinder (4)




- $P_1(\rho, \phi, z)$
- $P_2(\rho + \Delta\rho, \phi + \Delta\phi, z + \Delta z)$
- Panjang differensial $d\ell$

  $d\ell = d\rho\, \mathbf{a}_\rho + \rho\, d\phi\, \mathbf{a}_\phi + dz\, \mathbf{a}_z$
- Luas differensial

  $d\mathbf{s}_\rho = \rho\, d\phi\, dz\, \mathbf{a}_\rho$

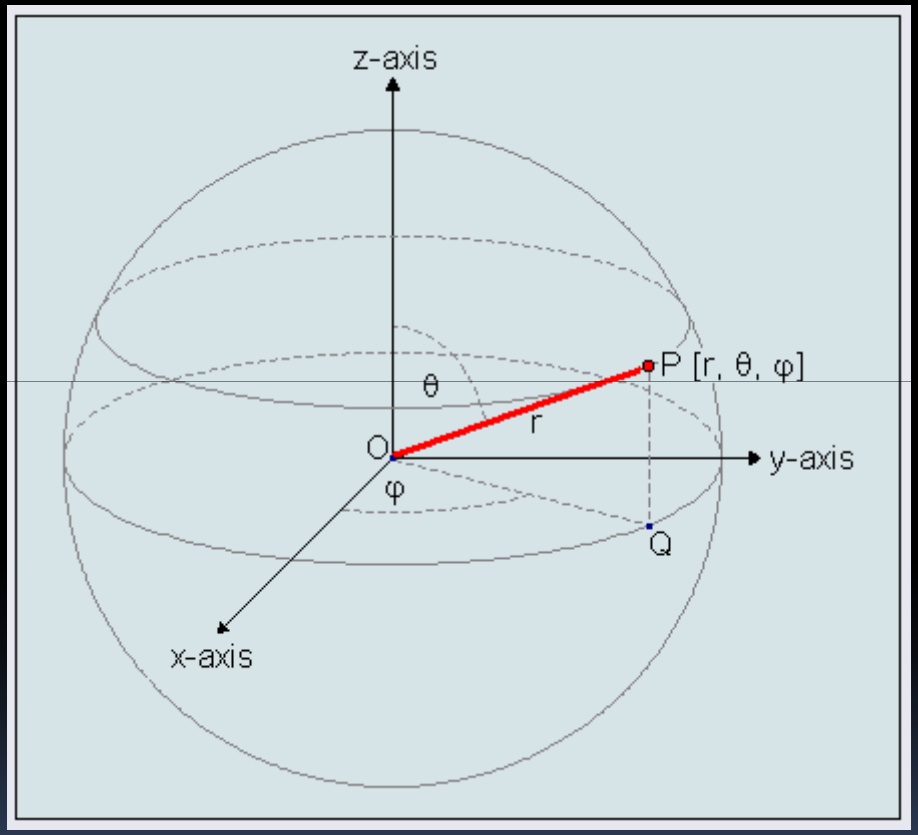
  $d\mathbf{s}_\phi = d\rho\, dz\, \mathbf{a}_\phi$

  $d\mathbf{s}_z = \rho\, d\rho\, d\phi\, \mathbf{a}_z$
- Volume differensial

  $dv = \rho\, d\rho\, d\phi\, dz$

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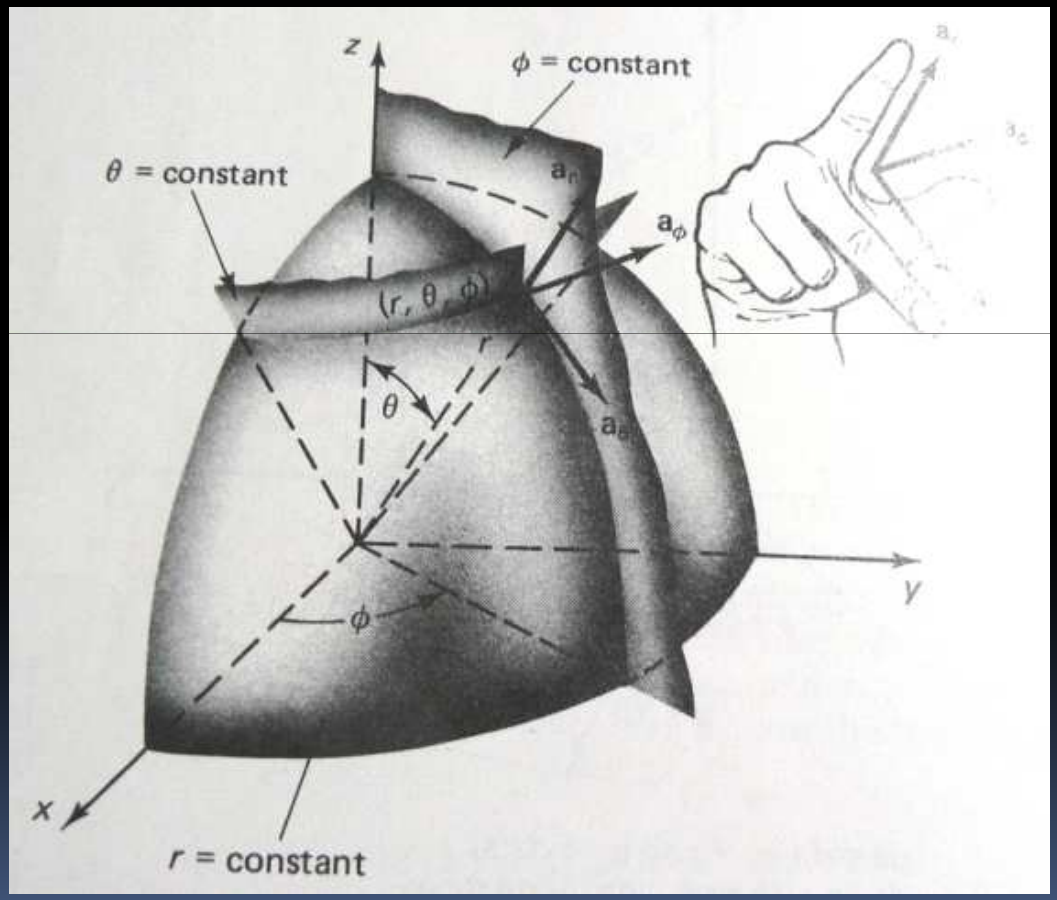
# Sistem koordinat bola (1)




- Sebuah titik P dinyatakan oleh P(r, θ, φ)

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# Sistem koordinat bola (2)

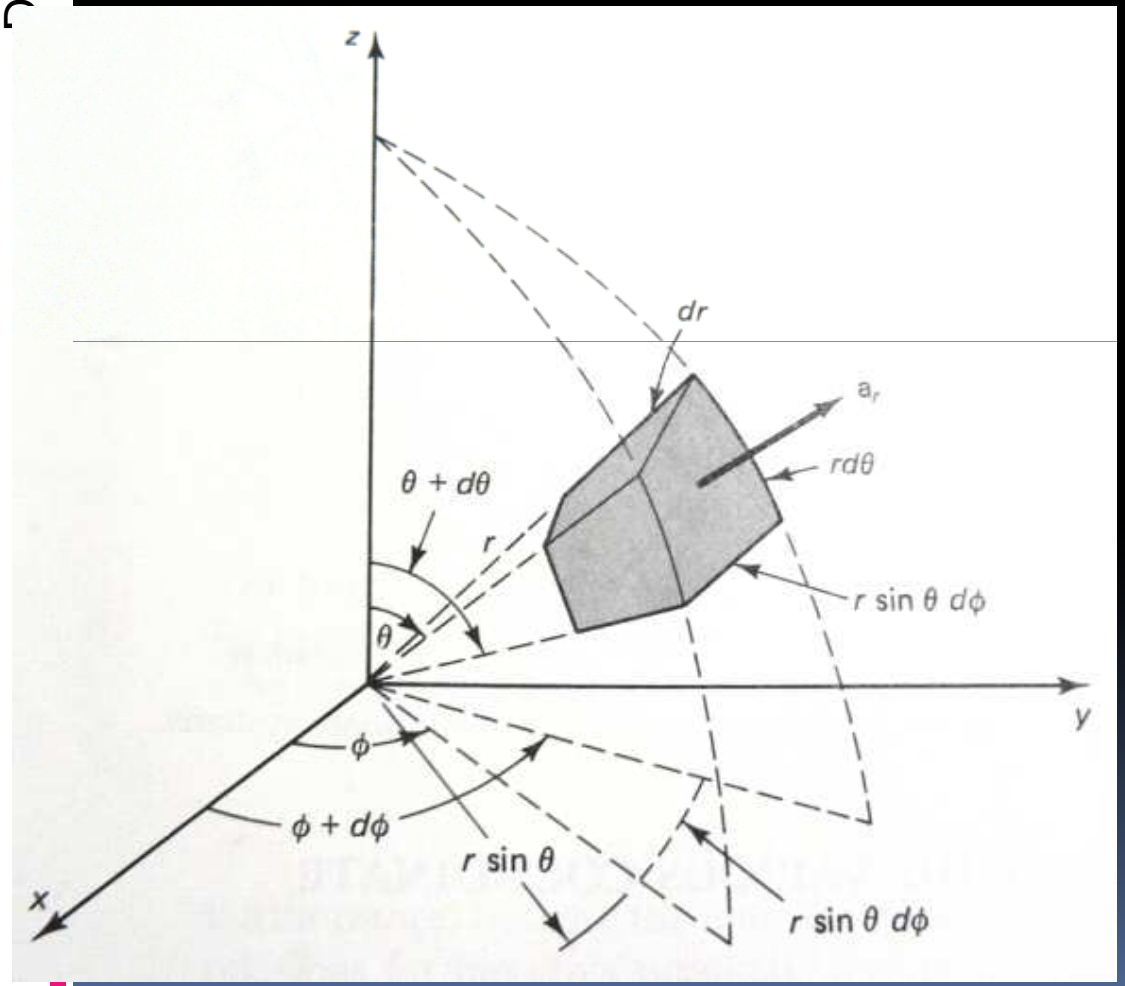



- Sebuah vektor **P** dinyatakan dengan :

$$\mathrm{P} = P_r \mathrm{a}_r + P_\theta \mathrm{a}_\theta + P_\phi \mathrm{a}_\phi$$

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# Sistem koordinat bola (3)




- $P_1(r, \theta, \phi)$
- $P_2(r + \Delta r, \theta + \Delta\theta, \phi + \Delta\phi)$
- Panjang differensial $d\ell$

  $d\ell = dr\, \mathbf{a}_r + r\, d\theta\, \mathbf{a}_\theta + r \sin\theta\, d\phi\, \mathbf{a}_\phi$
- Luas differensial

  $d\mathbf{s}_r = r^2 \sin\theta\, d\theta\, d\phi\, \mathbf{a}_r$

  $d\mathbf{s}_\theta = r \sin\theta\, dr\, d\phi\, \mathbf{a}_\theta$

  $d\mathbf{s}_\phi = r\, dr\, d\theta\, \mathbf{a}_\phi$
- Volume differensial

  $dv = r^2 \sin\theta\, dr\, d\theta\, d\phi$

# Transformasi Cartesian ke Silinder (2)

$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$$

$$\updownarrow$$

$$\mathbf{A} = A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z$$

???

$$A_\rho = \mathbf{A} \cdot \mathbf{a}_\rho = (A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z) \cdot \mathbf{a}_\rho$$

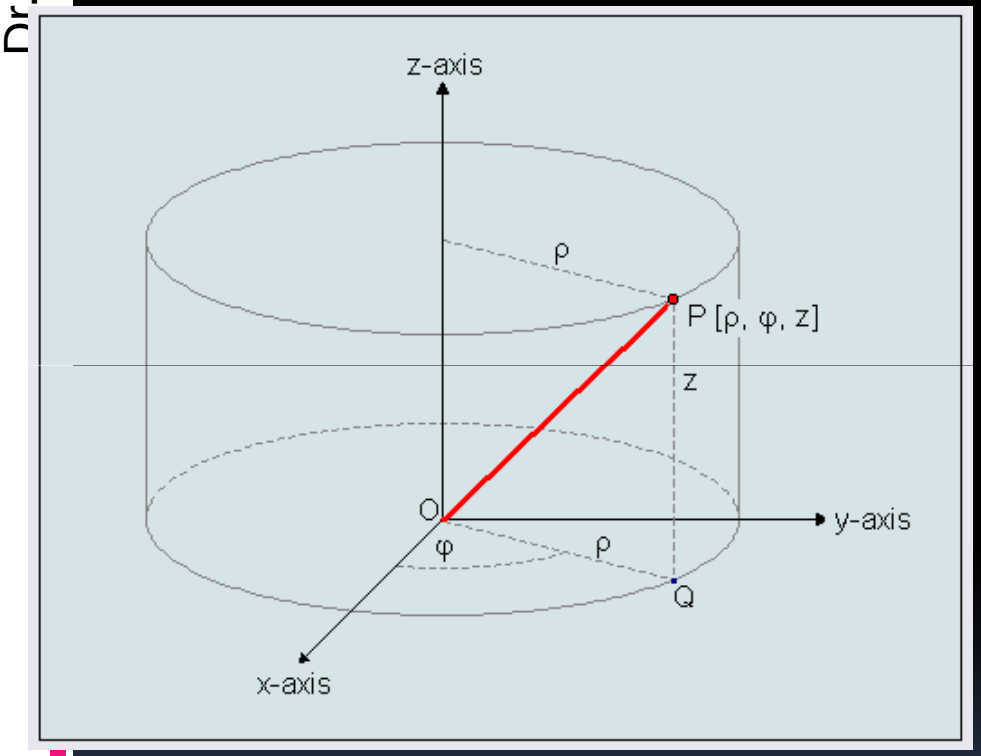
$$= A_x \mathbf{a}_x \cdot \mathbf{a}_\rho + A_y \mathbf{a}_y \cdot \mathbf{a}_\rho$$

$$A_\phi = \mathbf{A} \cdot \mathbf{a}_\phi = (A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z) \cdot \mathbf{a}_\phi$$

$$= A_x \mathbf{a}_x \cdot \mathbf{a}_\phi + A_y \mathbf{a}_y \cdot \mathbf{a}_\phi$$

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# Transformasi Cartesian ke Silinder (1)




- Car ke Silinder

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$

- Sil ke cartesian

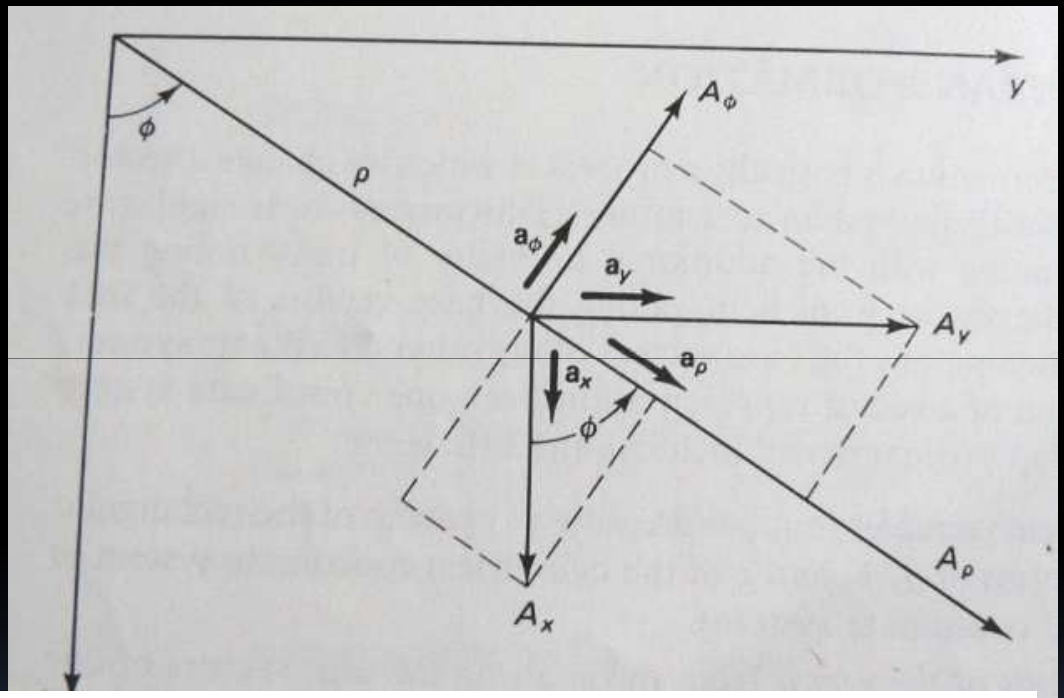
$$\rho = \sqrt{x^2 + y^2}$$

$$\varphi = \tan^{-1} \frac{y}{x}$$

$$z = z$$

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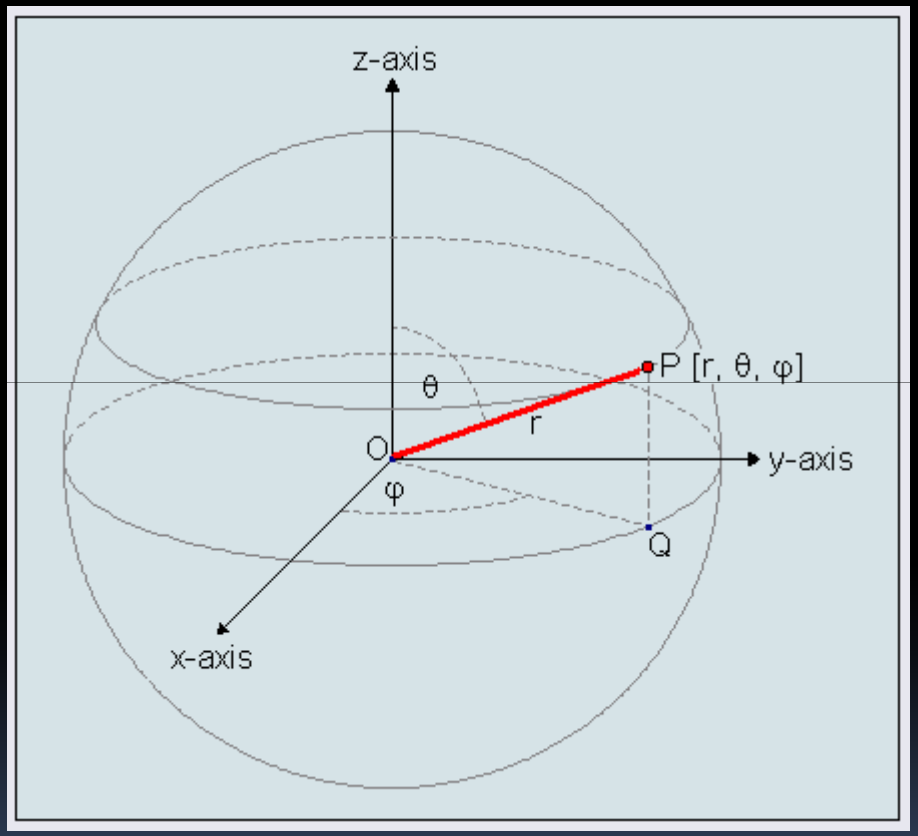
# Transformasi Cartesian ke Silinder (3)




| | $a_\rho$ | $a_\phi$ | $a_z$ |
|---|---|---|---|
| $a_x$ . | cos $\phi$ | - sin $\phi$ | 0 |
| $a_y$ . | sin $\phi$ | cos $\phi$ | 0 |
| $a_z$ . | 0 | 0 | 1 |

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# Transformasi Cartesian ke Bola (1)




- Car ke Bola

$$x = r \sin\theta \cos\phi$$

$$y = r \sin\theta \sin\phi$$

$$z = r \cos\theta$$

- Bola ke cartesian

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1} \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\varphi = \tan^{-1} \frac{y}{x}$$

# Transformasi Cartesian ke Bola (2)

$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$$



$$\mathbf{A} = A_r \mathbf{a}_r + A_\theta \mathbf{a}_\theta + A_\phi \mathbf{a}_\phi$$

???

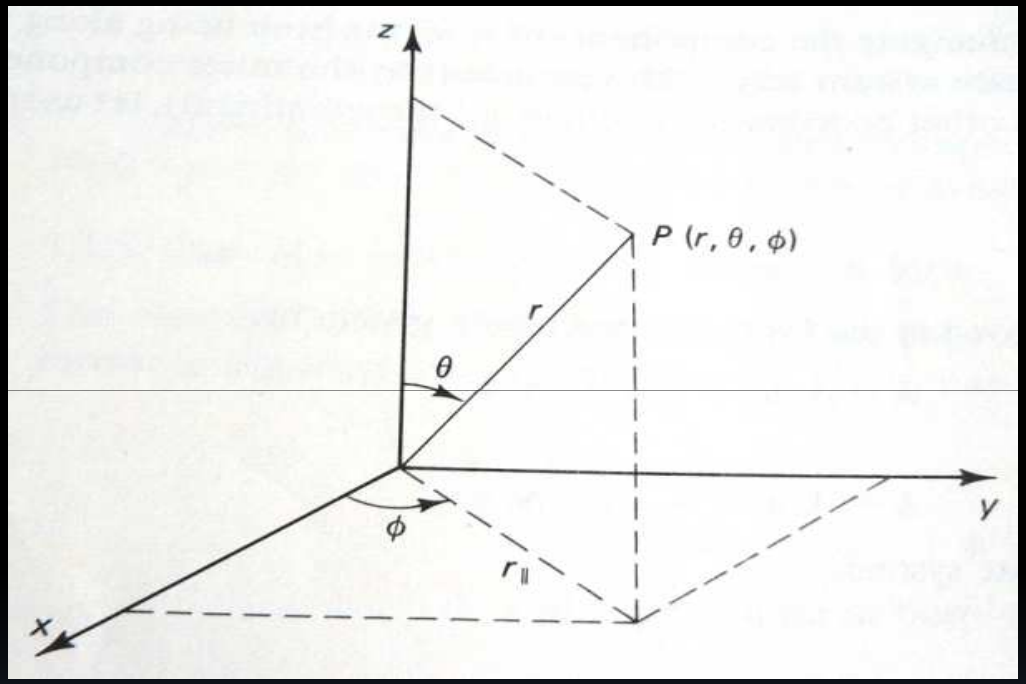
$$A_r = \mathbf{A} \cdot \mathbf{a}_r = A_x \mathbf{a}_x \cdot \mathbf{a}_r + A_y \mathbf{a}_y \cdot \mathbf{a}_r + A_z \mathbf{a}_z \cdot \mathbf{a}_r$$

$$A_\theta = \mathbf{A} \cdot \mathbf{a}_\theta = A_x \mathbf{a}_x \cdot \mathbf{a}_\theta + A_y \mathbf{a}_y \cdot \mathbf{a}_\theta + A_z \mathbf{a}_z \cdot \mathbf{a}_\theta$$

$$A_\phi = \mathbf{A} \cdot \mathbf{a}_\phi = A_x \mathbf{a}_x \cdot \mathbf{a}_\phi + A_y \mathbf{a}_y \cdot \mathbf{a}_\phi + A_z \mathbf{a}_z \cdot \mathbf{a}_\phi$$

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# Transformasi Cartesian ke Bola (3)




| | $\mathbf{a}_r$ | $\mathbf{a}_\theta$ | $\mathbf{a}_\phi$ |
|---|---|---|---|
| $\mathbf{a}_x$ . | $\sin\theta\cos\phi$ | $\cos\theta\cos\phi$ | $-\sin\phi$ |
| $\mathbf{a}_y$ . | $\sin\theta\sin\phi$ | $\cos\theta\sin\phi$ | $\cos\phi$ |
| $\mathbf{a}_z$ . | $\cos\theta$ | $-\sin\theta$ | 0 |